"Measure Maintainability of a Project"

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Submitted on 3rd December, 2025

Measure Maintainability of a Java Project

Install Prerequisites

Java JDK 8+: Download from https://adoptium.net/ .
Maven: Download from https://maven.apache.org/download.cgi.
Python 3+: Download from https://www.python.org.
Install javalang.

Build and Run the ck Tool for OO Metrics

1. Clone and build ck:
git clone https://github.com/mauricioaniche/ck.git
cd ck
mvn clean package -DskipTests

2. Run ck (generates class.csv)
java -jar "D:\8th\Software Maintenance\ck\target\ck-0.7.1-SNAPSHOT-jar-with-dependencies.jar" "D:\8th\Software Maintenance\File-Transfer-and-Chat-Project-in-Java\File Transfer and Chat Project in Java"

This will create a file named class.csv with 15 rows (13 real classes + 2 anonymous).

Class.csv

file	class	type	cbo	cboModified	fanin	fanout	wmc
D:\8th\Software M	clientform$Anonymous1	anonymous	0	0	0	0	1
D:\8th\Software M	chatform1	class	4	5	1	4	12
D:\8th\Software M	menu1$Anonymous1	anonymous	0	0	0	0	1
D:\8th\Software M	chatform	class	5	9	4	5	42
D:\8th\Software M	killall	class	1	2	1	1	7
D:\8th\Software M	filter	class	1	2	1	1	30
D:\8th\Software M	clientform	class	0	1	1	0	2
D:\8th\Software M	server3	class	3	6	3	3	17
D:\8th\Software M	chat	class	3	6	3	3	27
D:\8th\Software M	sendall	class	2	3	1	2	10
D:\8th\Software M	form1	class	1	2	1	1	5
D:\8th\Software M	mul2	class	3	5	2	3	24
D:\8th\Software M	menu1	class	3	4	1	3	13
D:\8th\Software M	quitting	class	0	1	1	0	5
D:\8th\Software M	ftpser1	class	1	1	0	1	1
			27	47			195
			Tcbo	TcboModified			Twmc

dit	noc	rfc	lcom	lcom*	tcc	lcc	totalMethodsQty	staticMethodsQty	publicMethodsQt
1	0	1	0	0	NaN	NaN	1	0	1
1	0	22	1	0.5925926	0.33333334	0.33333334	3	1	2
1	0	1	0	0	NaN	NaN	1	0	1
6	0	63	25	0.8214285	0.22222222	0.22222222	10	1	8
6	0	19	0	0.4	1	1	2	0	1
2	0	15	16	0.77777773	0.37878788	0.6818182	12	0	12
5	0	12	0	0	NaN	NaN	1	0	0
2	0	11	0	0.5555556	0.8055556	1	9	0	1
6	0	44	27	0.84210527	0.17777778	0.22222222	10	1	8
6	0	22	0	0.2	1	1	2	0	1
6	0	17	0	0.27272728	1	1	2	0	1
2	0	30	0	0.6000001	0.8	1	5	0	1
5	0	21	24	0.84615386	0.071428575	0.10714286	8	1	7
2	0	8	0	0.16666667	1	1	2	0	1
1	0	0	0	0	NaN	NaN	1	1	1
50	0						67		44
Tdit	Tnoc								

Compute Comments and Statements with Python Script

Here, the Python script named **fix.py** will create a CSV file, **extra_correct.csv**.

```
import os
import javalang

source_dir = r"D:\8th\Software Maintenance\File-Transfer-and-Chat-Project-in-Java\File Transfer and Chat Project in Java"
output_file = os.path.join(source_dir, "extra_correct.csv")

with open(output_file, "w", encoding="utf-8") as out:
    out.write("class,cl_comf,cl_stat\n")

    total_comf = 0.0
    total_stat = 0

    for root, _, files in os.walk(source_dir):
        for file in files:
            if file.endswith(".java"):
                path = os.path.join(root, file)
                try:
                    with open(path, "r", encoding="utf-8") as f:
                        lines = f.readlines()

                    comment_lines = 0
                    in_multi = False
                    for line in lines:
                        s = line.strip()
                        if in_multi:
```

```
                comment_lines += 1
                if "*/" in s: in_multi = False
                continue
            if s.startswith("//"):
                comment_lines += 1
            elif "/*" in s:
                comment_lines += 1
                if "*/" not in s: in_multi = True

        total_lines = sum(1 for line in lines if line.strip())
        comf = comment_lines / total_lines if total_lines > 0 else 0

        stmts = 0
        with open(path, "r", encoding="utf-8") as f:
            tree = javalang.parse.parse(f.read())
        for _, node in tree.filter(javalang.tree.Statement):
            stmts += 1

        class_name = file.replace(".java", "")
        out.write(f"{class_name},{round(comf,6)},{stmts}\n")

        total_comf += comf
        total_stat += stmts

    except:
        pass

out.write(f"\nTOTAL_COMF,,{round(total_comf,6)}\n")
out.write(f"TOTAL_STAT,,{total_stat}\n")

print("Done! File created → extra_correct.csv")
print(f"cl_comf total = {round(total_comf,6)}")
print(f"cl_stat total = {total_stat}")
```

The generated **extra_correct.csv** is

class	cl_comf	cl_stat
chat	0.074866	96
chatform	0.043478	185
chatform1	0.051852	61
clientform	0.121212	20
filter	0.010753	56
form1	0.055556	37
ftpser1	0.023166	128
killall	0.112676	36
menu1	0.010417	73
sendall	0.103774	66
TOTAL_COMF		0.607749
TOTAL_STAT		758

Extract Metrics from CSVs

Open class.csv in Excel. There are 15 rows, but use only 13 real classes (exclude anonymous ones like clientform$Anonymous1 and menu1$Anonymous1).

Sum these columns for the 13 real classes:

- cl_wmc = sum(wmc) = 195
- in_bases = sum(dit) = 50
- cu_cdused = sum(cbo) = 27
- cu_cdusers = sum(cboModified) = 47
- in_noc = sum(noc) = 0
- cl_func = sum(totalMethodsQty) = 67
- cl_func_pub = sum(publicMethodsQty) = 44
- cl_data = sum(totalFieldsQty) = 96
- cl_data_pub = sum(publicFieldsQty) = 0

From extra_correct.csv:

- cl_comf = TOTAL_COMF ≈ 0.607749
- cl_stat = TOTAL_STAT ≈ 758

Compute Maintainability

Maintainability = Analyzability + Changeability +Stability + Testability
Analyzability = cl_wmc + cl_comf +in_bases + cu_cdused
Changeability= cl_stat + cl_func + cl_data
Stability= cl_data publ+cu_cdusers+in_noc+cl_func_publ
Testability= cl_wmc + cl_func + cu_cdused

Analyzability	195 + 0.607749 + 50 + 27	272.6077
Changeability	758 + 67 + 96	921
Stability	0 + 47 + 0 + 44	91
Testability	195 + 67 + 27	289
Maintainability	272.6077 + 921 + 91 + 289	1573.6077

So, the Maintainability Score of the given project is **1573.6077**.